
Problem Set 2: Solutions

Econ 502: Advanced Microeconomics

Problem 1: Automation and Labor Demand

Part (a): Elasticity of substitution

The production function $Q = [0.6R^{0.5} + 0.4L^{0.5}]^2$ is a CES with parameter $\rho = 0.5$ (since the outer exponent is $1/\rho = 2$). The elasticity of substitution is:

$$\sigma = \frac{1}{1 - \rho} = \frac{1}{1 - 0.5} = 2$$

Since $\sigma = 2 > 1$, robots and workers are **good substitutes**—more substitutable than Cobb-Douglas ($\sigma = 1$). The firm can replace workers with robots relatively easily when the price of robots falls. A 1% decrease in the relative price of robots leads to a 2% increase in the robot-to-worker ratio.

Part (b): Cost-minimizing input mix for $Q = 1000$

The cost-minimizing condition requires $MRTS = r/w$. Computing the marginal products:

$$MP_R = 2[0.6R^{0.5} + 0.4L^{0.5}] \cdot 0.3R^{-0.5}, \quad MP_L = 2[0.6R^{0.5} + 0.4L^{0.5}] \cdot 0.2L^{-0.5}$$

Taking the ratio:

$$MRTS = \frac{MP_R}{MP_L} = \frac{0.3}{0.2} \cdot \frac{L^{0.5}}{R^{0.5}} = 1.5 \left(\frac{L}{R} \right)^{0.5}$$

Setting $MRTS = r/w$ and solving for the input ratio:

$$1.5 \left(\frac{L}{R} \right)^{0.5} = \frac{r}{w} \implies \frac{R}{L} = \left(\frac{1.5w}{r} \right)^2$$

With $r = 20$ and $w = 25$:

$$\frac{R}{L} = \left(\frac{1.5 \times 25}{20} \right)^2 = (1.875)^2 \approx 3.52$$

So $R \approx 3.52L$. Substituting into the production function with $Q = 1000$:

$$\sqrt{1000} = 0.6\sqrt{3.52} L^{0.5} + 0.4 L^{0.5} = L^{0.5}(0.6 \times 1.876 + 0.4) = 1.526 L^{0.5}$$

$$L^{0.5} = \frac{31.62}{1.526} = 20.72 \implies L \approx 429$$

$$R \approx 3.52 \times 429 \approx 1,510$$

$$R \approx 1,510, \quad L \approx 429$$

Total cost: $C = 20(1,510) + 25(429) = 30,200 + 10,725 = \$40,925$.

Part (c): Robot rental rate falls to $r = 10$

With the new prices:

$$\frac{R}{L} = \left(\frac{1.5 \times 25}{10} \right)^2 = (3.75)^2 = 14.06$$

So $R \approx 14.06L$. Substituting into the production constraint:

$$31.62 = L^{0.5}(0.6 \times 3.75 + 0.4) = 2.65 L^{0.5}$$

$$L^{0.5} = 11.93 \implies L \approx 142$$

$$R \approx 14.06 \times 142 \approx 1,997$$

$$R \approx 1,997, \quad L \approx 142$$

Percentage change in robot-to-worker ratio: The ratio goes from 3.52 to 14.06, an increase of $\frac{14.06-3.52}{3.52} \times 100 \approx 300\%$. The robot-to-worker ratio **quadruples**. This is because when r halves, the relative price w/r doubles, and with $\sigma = 2$ the input ratio responds by a factor of $2^2 = 4$.

The firm dramatically shifts toward robots: robot use rises from 1,510 to 1,997, while worker use falls from 429 to 142. The cheaper robots replace a large share of labor.

Part (d): Robot tax and job savings

A robot tax raising r back to 20 would restore the input mix from part (b). Comparing the two scenarios:

	$r = 10$ (no tax)	$r = 20$ (robot tax)	Change
Workers (L)	142	429	+287
Robots (R)	1,997	1,510	-\$487
Total cost	\$23,520	\$40,925	+\$17,405

The robot tax would **save approximately 287 worker-hours** per 1,000 units of output.

The trade-off: Production costs rise by about \$17,400 per 1,000 units (a 74% increase), which would be passed on to consumers through higher prices. The cost per worker-hour saved is approximately $\$17,400/287 \approx \61 . Whether this is worthwhile depends on how society values the protected jobs relative to the efficiency loss and higher consumer prices.

Problem 2: Profit Maximization and Taxation

A competitive firm maximizes profit $\pi = P \cdot q - C(q)$, yielding the first-order condition $P = MC(q)$.

Lump-sum tax on profits. With a lump-sum tax T , profit becomes $\pi = Pq - C(q) - T$. The FOC is still $P = MC(q)$ —the lump-sum tax does not depend on the level of output, so it does not affect the marginal conditions. The profit-maximizing quantity is **unchanged**. (The firm earns less profit, but it cannot reduce its tax by producing more or less.)

Proportional tax on profits. With tax rate t , after-tax profit is $\pi = (1 - t)[Pq - C(q)]$. The FOC is $(1 - t)[P - MC(q)] = 0$. Since $(1 - t) > 0$, this gives $P = MC(q)$, the same condition as before. The profit-maximizing quantity is **unchanged**. (Intuitively, if q^* maximizes pre-tax profit, it also maximizes $(1 - t)$ times pre-tax profit.)

Tax on each unit of output. With a per-unit tax τ , profit becomes $\pi = Pq - C(q) - \tau q = (P - \tau)q - C(q)$. The FOC is $P - \tau = MC(q)$, so the firm produces where marginal cost equals the net price $P - \tau$. Since MC is increasing, the profit-maximizing quantity **decreases**. The per-unit tax is equivalent to a reduction in the output price—it shifts the effective price down and reduces output.

Tax on labor input. If the firm uses labor L at wage w and the government imposes a tax t_L per unit of labor, the effective wage rises to $w + t_L$. This increases the cost function $C(q)$, which shifts $MC(q)$ upward. Since the firm produces where $P = MC(q)$, a higher MC curve means a lower profit-maximizing quantity. Output **decreases**. The firm also substitutes away from labor toward other inputs.

Lump-sum tax: no effect. Proportional tax: no effect. Output tax: q falls. Labor tax: q falls.

Problem 3: Trade Policy and the Portable Radio Market

Part (a): Domestic equilibrium

Setting domestic demand equal to domestic supply:

$$D(P) = S(P) \implies 5,000 - 100P = 150P \implies 250P = 5,000$$

$$P^* = \$20, \quad Q^* = 3,000 \text{ thousand radios}$$

Part (b): Free trade at world price $P_W = \$10$

At $P = 10$:

- Domestic demand: $Q_D = 5,000 - 100(10) = 4,000$ thousand
- Domestic supply: $Q_S = 150(10) = 1,500$ thousand

$$\text{Imports} = Q_D - Q_S = 4,000 - 1,500 = 2,500 \text{ thousand radios}$$

The price falls from \$20 to \$10, domestic consumption rises, domestic production falls, and the gap is filled by imports.

Part (c): \$5 tariff

With the tariff, the domestic price rises to $P = 10 + 5 = \$15$:

- Domestic demand: $Q_D = 5,000 - 100(15) = 3,500$ thousand
- Domestic supply: $Q_S = 150(15) = 2,250$ thousand
- Imports: $3,500 - 2,250 = 1,250$ thousand

Tariff revenue:

$$R = \$5 \times 1,250 = \$6,250 \text{ thousand}$$

Change in consumer surplus (trapezoid between $P = 10$ and $P = 15$ under demand curve):

$$\Delta CS = -\frac{1}{2}(Q_D^{old} + Q_D^{new})(P^{new} - P^{old}) = -\frac{1}{2}(4,000 + 3,500)(5) = -\$18,750 \text{ thousand}$$

Consumer surplus transferred to domestic producers (increase in producer surplus—trapezoid between $P = 10$ and $P = 15$ under supply curve):

$$\Delta PS = \frac{1}{2}(Q_S^{old} + Q_S^{new})(P^{new} - P^{old}) = \frac{1}{2}(1,500 + 2,250)(5) = \$9,375 \text{ thousand}$$

Deadweight loss (the CS lost that is neither captured by producers nor by the government):

$$DWL = |\Delta CS| - \Delta PS - R = 18,750 - 9,375 - 6,250 = \$3,125 \text{ thousand}$$

This DWL consists of two triangles:

- *Production inefficiency*: $\frac{1}{2}(5)(2,250 - 1,500) = 1,875$ (resources wasted producing domestically what could be imported more cheaply)
- *Consumption loss*: $\frac{1}{2}(5)(4,000 - 3,500) = 1,250$ (lost surplus from consumers priced out of the market)

Part (d): Voluntary Export Restraint (VER) of 1,250 thousand

With the VER limiting imports to 1,250 thousand, the domestic price must adjust so that $Q_D - Q_S = 1,250$:

$$5,000 - 100P - 150P = 1,250 \implies P = \$15$$

The equilibrium price, domestic demand, domestic supply, and import quantity are **identical** to the tariff case.

Key difference: With a tariff, the government collects \$6,250 thousand in tariff revenue. With the VER, there is no tariff—foreign exporters sell at the higher domestic price of \$15 instead of the world price of \$10, so they capture the \$6,250 thousand as **quota rents**.

From the domestic country's perspective, the VER is strictly worse than the tariff:

	Tariff	VER
CS loss	\$-\$18,750	\$-\$18,750
PS gain	+9,375	+9,375
Gov. revenue	+6,250	0
Quota rents to foreigners	0	\$-\$6,250
Net domestic welfare change	-3,125	-9,375

The VER transfers an additional \$6,250 thousand to foreign producers, making the domestic welfare loss three times larger than the tariff.

Problem 4: General Equilibrium Exchange

Part (a): MRS at initial endowment

For $U = X \cdot Y$ (Cobb-Douglas with equal exponents):

$$MRS = \frac{MU_X}{MU_Y} = \frac{Y}{X}$$

At the initial endowment $(X_A, Y_A) = (70, 30)$ and $(X_B, Y_B) = (30, 70)$:

$$MRS_A = \frac{30}{70} = \frac{3}{7} \approx 0.43, \quad MRS_B = \frac{70}{30} = \frac{7}{3} \approx 2.33$$

Since $MRS_A \neq MRS_B$, the initial allocation is **not Pareto efficient**. There are gains from trade: Ben values X much more highly than Ana does (he is willing to give up 7/3 units of Y for one unit of X, while Ana would only give up 3/7). Ana should trade X to Ben in exchange for Y.

Part (b): Competitive equilibrium at $P_X/P_Y = 1$

Let $P_Y = 1$ (numeraire), so $P_X = 1$.

Ana's problem: Budget = $P_X \cdot 70 + P_Y \cdot 30 = 100$. With $U = XY$, the demand functions are:

$$X_A = \frac{I_A}{2P_X} = \frac{100}{2} = 50, \quad Y_A = \frac{I_A}{2P_Y} = \frac{100}{2} = 50$$

Ben's problem: Budget = $P_X \cdot 30 + P_Y \cdot 70 = 100$. By the same logic:

$$X_B = \frac{100}{2} = 50, \quad Y_B = \frac{100}{2} = 50$$

Market clearing: Total demand for X: $50 + 50 = 100 =$ total endowment.

$$(X_A, Y_A) = (50, 50), \quad (X_B, Y_B) = (50, 50)$$

Ana trades 20 units of X to Ben in exchange for 20 units of Y.

Part (c): Verifying Pareto efficiency

At the equilibrium allocation:

$$MRS_A = \frac{Y_A}{X_A} = \frac{50}{50} = 1, \quad MRS_B = \frac{Y_B}{X_B} = \frac{50}{50} = 1$$

$$MRS_A = MRS_B = 1 = \frac{P_X}{P_Y}$$

The competitive equilibrium is Pareto efficient: both consumers' marginal rates of substitution are equal (to each other and to the price ratio), so there is no way to make one person better off without making the other worse off.

Part (d): Contract curve and the First Welfare Theorem

The **contract curve** is the set of all Pareto efficient allocations, where $MRS_A = MRS_B$:

$$\frac{Y_A}{X_A} = \frac{Y_B}{X_B} = \frac{100 - Y_A}{100 - X_A}$$

Cross-multiplying: $Y_A(100 - X_A) = X_A(100 - Y_A)$, which simplifies to:

$$Y_A = X_A$$

The contract curve is the diagonal of the Edgeworth box (equivalently, $Y_B = X_B$). Every Pareto efficient allocation gives each person equal amounts of both goods.

The **First Welfare Theorem** states that every competitive equilibrium allocation is Pareto efficient. We verified this in part (c): the equilibrium allocation (50, 50) for both consumers lies on the contract curve ($Y_A = X_A = 50$). The competitive market, guided only by prices, achieves an efficient outcome without any central planner.

Problem 5: Optimal Redistribution with Distortionary Taxation

Part (a): Deriving the optimal tax rate

Substituting the consumption expressions into the welfare function:

$$c_P = y_P + \tau y_R - \frac{1}{2}k\tau^2 y_R, \quad c_R = (1 - \tau)y_R$$

$$W(\tau) = \alpha \left[y_P + \tau y_R - \frac{1}{2}k\tau^2 y_R \right] + (1 - \alpha)(1 - \tau)y_R$$

Expanding:

$$W(\tau) = \alpha y_P + \alpha \tau y_R - \frac{1}{2}\alpha k \tau^2 y_R + (1 - \alpha)y_R - (1 - \alpha)\tau y_R$$

Taking the first-order condition:

$$\frac{dW}{d\tau} = \alpha y_R - \alpha k \tau y_R - (1 - \alpha)y_R = 0$$

Dividing through by y_R :

$$\alpha - \alpha k \tau - (1 - \alpha) = 0$$

$$2\alpha - 1 = \alpha k \tau$$

$$\tau^* = \frac{2\alpha - 1}{\alpha k}$$

The second-order condition confirms this is a maximum: $\frac{d^2W}{d\tau^2} = -\alpha k y_R < 0$. Note that $\tau^* > 0$ requires $\alpha > 1/2$: positive redistribution is only optimal when society places more weight on the poor than on the rich.

Part (b): Comparative statics

Rewriting τ^* in a more convenient form:

$$\tau^* = \frac{2}{k} - \frac{1}{\alpha k}$$

(i) Effect of inequality aversion α :

$$\frac{\partial \tau^*}{\partial \alpha} = \frac{1}{\alpha^2 k} > 0$$

Higher inequality aversion (larger α) increases the optimal tax rate. **Intuition:** When society cares more about the well-being of the poor relative to the rich, it is willing to tolerate more deadweight

loss in order to transfer additional resources to the poor. A more egalitarian social welfare function justifies a larger tax-and-transfer program.

(ii) Effect of the distortion parameter k :

$$\frac{\partial \tau^*}{\partial k} = -\frac{2\alpha - 1}{\alpha k^2} < 0 \quad (\text{for } \alpha > 1/2)$$

A higher distortion cost (larger k) decreases the optimal tax rate. **Intuition:** When taxation is more distortionary—meaning more revenue is lost to deadweight loss per dollar collected—the government should tax less aggressively. The “leaky bucket” of redistribution becomes leakier: each dollar taken from the rich delivers less than a dollar to the poor, so the marginal cost of redistribution rises and the optimal level falls.